Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The reaction $\mathrm{A(g) + 2\ B(\ g) \rightarrow C(g) + D(g)}$ is an elementary process. In an experiment involving this reaction, the initial partial pressures of A and B are $\mathrm{p_A = 0.60\ atm}$ and $\mathrm{p_B = 0.80}$ atm respectively. When $\mathrm{p_C = 0.20\ atm}$, the rate of the reaction relative to the initial rate is

(A) $1/6$

(B) $1/12$

(C) $1/36$

(D) $1/18$

Q.2

Consider the reaction :

$\mathrm{Cl_2(aq) + H_2\ S(aq) \longrightarrow S(s) + 2H^+(aq) + 2Cl^-(aq)}$

The rate equation for this reaction is

rate $= \mathrm{k\ [Cl_2]\ [H_2\ S]}$

Which of these mechanisms is/are consistent with this rate equation?

A. $\mathrm{Cl_2 + H_2\ S \rightarrow H^+ + Cl^- + Cl^+ + HS^-}$ (slow)

$\mathrm{Cl^+ + HS^- \rightarrow H^+ + Cl^- + S}$ (fast)

B. $\mathrm{H_2\ S \rightleftharpoons H^+ + HS^-}$ (fast equilibrium)

$\mathrm{Cl_2 + HS^- \rightarrow 2Cl^- + H^+ + S}$ (Slow)

(A) B only

(B) Both A and B

(C) Neither A nor B

(D) A only

Q.3

The rate law for a reaction between the substances A and B is given by Rate $= \mathrm{k[A]^n[B]^m}$

On doubling the concentration of A and halving the concentration of B, the ratio of the new rate to the earlier rate of the reaction will be as

(A) $\mathrm{(m + n)}$

(B) $\mathrm{(n - m)}$

(C) $2^{\mathrm{(n-m)}}$

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(D) $\frac{1}{2^{(m+n)}}$

Q.4

Consider the reaction
$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$
The equality relationship between $\frac{d[NH_3]}{dt}$ and $-\frac{d[H_2]}{dt}$ is

(A) $+\frac{d[NH_3]}{dt} = -\frac{2}{3}\frac{d[H_2]}{dt}$

(B) $+\frac{d[NH_3]}{dt} = -\frac{3}{2}\frac{d[H_2]}{dt}$

(C) $\frac{d[NH_3]}{dt} = -\frac{d[H_2]}{dt}$

(D) $\frac{d[NH_3]}{dt} = -\frac{1}{3}\frac{d[H_2]}{dt}$

Q.5

For the reaction
$N_2O_5(g) \longrightarrow 2NO_2(g) + 1/2O_2(g)$ the value
of rate of disappearance of N_2O_5 is given as $6.25 \times 10^{-3}\ mol\ L^{-1}\ s^{-1}$. The rate of formation ofNO_2 and O_2 is given respectively as :

(A) $6.25 \times 10^{-3}\ mol\ L^{-1}\ s^{-1}$ and
$6.25 \times 10^{-3}\ mol\ L^{-1}\ s^{-1}$

(B) $1.25 \times 10^{-2}\ mol\ L^{-1}\ s^{-1}$ and
$3.125 \times 10^{-3}\ mol\ L^{-1}\ s^{-1}$

(C) $6.25 \times 10^{-3}\ mol\ L^{-1}\ s^{-1}$ and
$3.125 \times 10^{-3}\ mol\ L^{-1}\ s^{-1}$

(D) $1.25 \times 10^{-2}\ mol\ L^{-1}\ s^{-1}$ and
$6.25 \times 10^{-3}\ mol\ L^{-1}\ s^{-1}$

Q.6

The reactions rate $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$ was measured $\frac{d[NH_3]}{dt} = 2 \times 10^{-4}\ molsec^{-1}$ The rates of reactions expressed in terms of N_2 and H_2 are

(A) Rate in terms of $\mathbf{N_2} = 2 \times 10^{-4}\ (molL^{-1}sec^{-1})$
Rate in terms of $\mathbf{H_2} = 2 \times 10^{-4}\ (molL^{-1}sec^{-1})$

(B) Rate in terms of $\mathbf{N_2} = 3 \times 10^{-4}\ (molL^{-1}sec^{-1})$
Rate in terms of $\mathbf{H_2} = 1 \times 10^{-4}\ (molL^{-1}sec^{-1})$

(C) Rate in terms of $\mathbf{N_2} = 1 \times 10^{-4}\ (molL^{-1}sec^{-1})$
Rate in terms of $\mathbf{H_2} = 3 \times 10^{-4}\ (molL^{-1}sec^{-1})$

(D) Rate in terms of $\mathbf{N_2} = 2 \times 10^{-1}\ (molL^{-1}sec^{-1})$
Rate in terms of $\mathbf{H_2} = 2 \times 10^{-3}\ (molL^{-1}sec^{-1})$

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Q.7

A following mechanism has been proposed for reaction $2\ \mathrm{A} + \mathrm{B} \rightarrow \mathrm{D} + \mathrm{E}$
$\mathrm{A} + \mathrm{B} \rightarrow \mathrm{C} + \mathrm{D}(\text{ slow })$
$\mathrm{A} + \mathrm{C} \rightarrow \mathrm{E}$ (fast)
The rate law expression for the reaction is:

(A) $\mathrm{r} = \mathrm{k}[\mathrm{A}]^2[\ \mathrm{B}]$

(B) $\mathrm{r} = \mathrm{k}[\mathrm{A}][\mathrm{B}]$

(C) $\mathrm{r} = \mathrm{k}[\mathrm{A}]^2$

(D) $\mathrm{r} = \mathrm{k}[\mathrm{A}][\mathrm{C}]$

Q.8

The rate law for the reaction $2\mathrm{X} + \mathrm{Y} \rightarrow \mathrm{Z}$ is Rate $= k[\mathrm{X}][\mathrm{Y}]$. The correct statement with regard to this relation is

(A) the unit of k is s^{-1}

(B) the rate of the reaction is independent of $[\mathrm{X}]$ and $[\mathrm{Y}]$

(C) for this reaction $t_{1/2}$ is independent of initial concentrations of reactant

(D) the rate of formation of Z is half of the rate of disappearance of X

Q.9

In a reaction, $2\ \mathrm{A} \rightarrow$ products, the concentration of A decreases from $0.50\mathrm{M}$ to $0.38\mathrm{M}$ in $10\ \mathrm{min}$. What is the rate of disappearance of A (in Ms^{-1}) during this interval?

(A) 0.012

(B) 0.024

(C) 2×10^{-3}

(D) 2×10^{-4}

Q.10

A chemical reaction is catalyzed by a catalyst X Hence X

(A) reduces enthalpy of the reaction

(B) decreases rate constant of the reaction

(C) increases activation energy of the reaction

(D) does not affect equilibrium constant of the reaction

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Q.11

A catalyst is a substance which :

(A) is always in the same phase as in the reaction

(B) alters the equilibrium in a reaction

(C) does not participate in the reaction but alters the rate of reaction

(D) participates in the reaction and provides an easier pathway for the same

Q.12

The inversion of cane sugar proceeds with half-life of 500 minute at $\mathrm{pH}=5$ for any concentration of sugar. However if $\mathrm{pH}=6$, the half-life changes to 50 minute. The rate law expression for the sugar inversion can be written as

(A) $\mathrm{r}=\mathrm{K}[\text{ sugar }]^{2}[\mathrm{H}]^{6}$

(B) $\mathrm{r}=\mathrm{K}[\text{ sugar }]^{1}[\mathrm{H}+]^{0}$

(C) $\mathrm{r}=\mathrm{K}[\text{ sugar }]^{0}\left[\mathrm{H}^{+}\right]^{6}$

(D) $\mathrm{r}=\mathrm{K}[\text{ sugar }]^{0}\left[\mathrm{H}^{+}\right]^{1}$

Q.13

The kinetic datas for the reaction:
$2\ \mathrm{A}+\mathrm{B}_{2} \rightarrow 2\mathrm{AB}$ are as given below:
\begin{array}{|l|l|l|}
\hline[\mathrm{A}] \mathrm{mol} \mathrm{L}^{-1} & {\left[\mathrm{~B}_{2}\right] \mathrm{mol} \mathrm{L}^{-1}} & Rate \mathrm{mol} \mathrm{L}^{-1} \mathrm{~min}^{-1} \\
\hline 0.5 & 1.0 & 2.5 \times 10^{-3} \\
\hline 1.0 & 1.0 & 5.0 \times 10^{-3} \\
\hline 0.5 & 2.0 & 1 \times 10^{-2} \\
\hline
\end{array}
Hence the order of reaction w.r.t. A and B_{2} are, respectively,

(A) 1 and 2

(B) 2 and 1

(C) 1 and 1

(D) 2 and 2

Q.14

Reaction rates are determined by all of the following factors except:

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(A) the number of collisions between molecules

(B) the force of collision between molecules

(C) the orientation of collision between molecules

(D) the spontaneity of the reaction

Q.15

For a reaction $\frac{1}{2}\ \mathrm{A} \rightarrow 2\ \mathrm{B}$, rate of disappearance of ' A' is related to the rate of appearance of 'B' by the expression

(A) $-\frac{\mathrm{d}[\mathrm{A}]}{\mathrm{dt}}=\frac{1}{4}\frac{\mathrm{d}[\mathrm{B}]}{\mathrm{dt}}$

(B) $-\frac{\mathrm{d}[\mathrm{A}]}{\mathrm{dt}}=\frac{\mathrm{d}[\mathrm{B}]}{\mathrm{dt}}$

(C) $-\frac{\mathrm{d}[\mathrm{A}]}{\mathrm{dt}}=4\frac{\mathrm{d}[\mathrm{B}]}{\mathrm{dt}}$

(D) $-\frac{\mathrm{d}[\mathrm{A}]}{\mathrm{dt}}=\frac{1}{2}\frac{\mathrm{d}[\mathrm{B}]}{\mathrm{dt}}$

Q.16

The terms rate of reaction and rate of appearance (or disappearance) of reactant (or product)

(A) Represent one and the same physical quantity

(B) Differ by constant factor

(C) Are positive parameters and have same value

(D) May or may not have same value depending upon the stoichiometric coefficient of reactant (or product) in the balanced chemical equation

Q.17

For the reaction $3\mathrm{BrO}^{-} \rightarrow \mathrm{BrO}_3^{-}+2\mathrm{Br}$ in alkaline aqueous solution, the value of the second order (in BrO^{-}) rate constant at $80^{\circ}\mathrm{C}$ in the rate law $-\Delta\left[\mathrm{BrO}^{-}\right]/\Delta \mathrm{t}$ was found to be $0.056\ \mathrm{L\ mol}^{-1}\ \mathrm{s}^{-1}$. Rate constant when the rate law is written for $\frac{\Delta\left[\mathrm{BrO}_3^{-}\right]}{\Delta \mathrm{t}}$ is

(A) $0.056\ \mathrm{L\ mol}^{-1}\ \mathrm{s}^{-1}$

(B) $\frac{0.056}{3}\ \mathrm{L\ mol}^{-1}\ \mathrm{s}^{-1}$

(C) $0.056 \times \frac{2}{3}\ \mathrm{L\ mol}^{-1}\ \mathrm{s}^{-1}$

(D) $0.028\ \mathrm{L\ mol}^{-1}\ \mathrm{s}^{-1}$

Q.18

For the reaction $\mathrm{A(g)} \rightarrow \mathrm{B(g)}+\mathrm{C(g)}$

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rate law is $-\frac{\mathrm{d}[\mathrm{A}]}{\mathrm{dt}}=\mathrm{k}[\mathrm{A}]$
At the start pressure is $100\ \mathrm{mm}$ and after 10 min, pressure is $120\ \mathrm{mm}$, hence, rate constant $\left(\mathrm{min}^{-1}\right)$ is

(A) $\frac{2.303}{10}\log\frac{6}{5}$

(B) $\frac{2.303}{10}\log 5$

(C) $\frac{2.303}{10}\log\frac{5}{4}$

(D) $\frac{2.303}{10}\log\frac{5}{6}$

Q.19

A G.M. counter is used to study the radioactive process of first-order. In the absence of radioactive substance A, it counts 3 disintegration per second (dps). When A is placed in the G.M. counter, it records 23 dps at the start and 13 dps after 10 minutes. It records x dps after next 10 minutes and A has half-life period y minutes. x and y are

(A) $8\mathrm{dps}, 10\ \mathrm{min}$

(B) $5\mathrm{dps}, 10\ \mathrm{min}$

(C) $5\mathrm{dps}, 20\ \mathrm{min}$

(D) $5\mathrm{dps}, 5\ \mathrm{min}$

Q.20

For the reaction. $\mathrm{A}+2\mathrm{B}\longrightarrow\mathrm{C}$, the reaction rate is doubled if the concentration of A is doubled. The rate is increased by four times when the concentrations of both A and B are increased by four times. The order of the reaction is.....

(A) 4

(B) 3

(C) 1

(D) 2

Answers & Solutions

Q.1 Answer:

$1/6$

Solution:

$\mathrm{A(g)}+2\,\mathrm{B(\ g)}\longrightarrow\mathrm{C(g)}+\mathrm{D(g)}$
Rate $=\mathrm{k}[\mathrm{A}][\mathrm{B}]^2=\mathrm{k}(0.60)(0.80)^2$ **...(i)**
when $\mathrm{p_c}=0.20\ \mathrm{atm}$ $\mathrm{p_A}$ **is reduced to** 0.40 **and**
$\mathrm{p_B}=0.40$ **(See stoichiometric representation)**

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Rate $= \mathrm{k}[0.40][0.40]^2$ **...(ii)**

∴ (ii) divide by (i) $= \frac{0.40 \times 0.40 \times 0.40}{0.60 \times 0.80 \times 0.80} = \frac{1}{6}$

Q.2 Answer:

A only

Solution:

Since the slow step is the rate determining step hence if we consider option (A) we find

Rate $= k\left[\mathrm{Cl}_2\right]\left[\mathrm{H}_2\ \mathrm{S}\right]$

Now if we consider option (B) we find

Rate $= k\left[\mathrm{Cl}_2\right]\left[\mathrm{HS}^-\right]$ **...(i)**

For equation,

$\mathrm{H}_2\ \mathrm{S} \rightleftharpoons \mathrm{H}^+ + \mathrm{HS}^-$

$K = \frac{\left[\mathrm{H}^+\right]\left[\mathrm{HS}^-\right]}{\mathrm{H}_2\ \mathrm{S}}$

or $\left[\mathrm{HS}^-\right] = \frac{K[\mathrm{H}_2\ \mathrm{S}]}{\mathrm{H}^+}$

Substituting this value in equation (i) we find

Rate $= k\left[\mathrm{Cl}_2\right] K \frac{[\mathrm{H}_2\ \mathrm{S}]}{\mathrm{H}^+} = k' \frac{[\mathrm{Cl}_2][\mathrm{H}_2\ \mathrm{S}]}{[\mathrm{H}^+]}$

Thus slow step should involve 1 molecule of Cl_2 **and 1 molecule of** $\mathrm{H}_2\ \mathrm{S}$**.**

hence only, mechanism (A) is consistent with the given rate equation.

Q.3 Answer:

$2^{(\mathrm{n}-\mathrm{m})}$

Solution:

Rate $_1 = \mathrm{k}[\mathrm{A}]^{\mathrm{n}}[\mathrm{B}]^{\mathrm{m}}$; **Rate** $_2 = \mathrm{k}[2\ \mathrm{A}]^{\mathrm{n}}[1/2\ \mathrm{B}]^{\mathrm{m}}$

$$\therefore \frac{\text{Rate }_2}{\text{Rate }_1} = \frac{k[2A]^n[1/2B]^m}{k[A]^n[B]^m}$$

$= [2]^n[1/2]^m = 2^n \cdot 2^{-m} = 2^{n-m}$

Q.4 Answer:

$+\frac{\mathrm{d}[\mathrm{NH}_3]}{\mathrm{dt}} = -\frac{2}{3}\frac{\mathrm{d}[\mathrm{H}_2]}{\mathrm{dt}}$

Solution:

If we write rate of reaction in terms of concentration of

NH_3 **and** H_2**, then**

Rate of reaction $= \frac{1}{2}\frac{\mathrm{d}[\mathrm{NH}_3]}{\mathrm{dt}} = -\frac{1}{3}\frac{\mathrm{d}[\mathrm{H}_2]}{\mathrm{dt}}$

So, $\frac{\mathrm{d}[\mathrm{NH}_3]}{\mathrm{dt}} = -\frac{2}{3}\frac{\mathrm{d}[\mathrm{H}_2]}{\mathrm{dt}}$

Q.5 Answer:

$1.25 \times 10^{-2}\ \mathrm{mol\ L^{-1}\ s^{-1}}$ and
$3.125 \times 10^{-3}\ \mathrm{mol\ L^{-1}\ s^{-1}}$

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Solution:

$\mathrm{N_2O_5(g) \longrightarrow 2NO_2(g) + 1/2O_2(g)}$
$-\frac{\mathrm{d}}{\mathrm{dt}}[\mathrm{N_2O_5}] = +\frac{1}{2}\frac{\mathrm{d}}{\mathrm{dt}}[\mathrm{NO_2}] = 2\frac{\mathrm{d}}{\mathrm{dt}}[\mathrm{O_2}]$
$\frac{\mathrm{d}}{\mathrm{dt}}[\mathrm{NO_2}] = 1.25 \times 10^{-2}\ \mathrm{mol\ L^{-1}\ s^{-1}}$ and
$\frac{\mathrm{d}}{\mathrm{dt}}[\mathrm{O_2}] = 3.125 \times 10^{-3}\ \mathrm{mol\ L^{-1}\ s^{-1}}$

Q.6 Answer:

Rate in terms of $\mathbf{N_2} = 1 \times 10^{-4}\ \left(\mathrm{molL^{-1}sec^{-1}}\right)$
Rate in terms of $\mathbf{H_2} = 3 \times 10^{-4}\ \left(\mathrm{molL^{-1}sec^{-1}}\right)$

Solution:

$\mathrm{N_2 + 3H_2 \rightleftharpoons 2NH_3}$; **Rate is given by any of the expressions**

$\frac{-\mathrm{d[N_2]}}{\mathrm{dt}} = -\frac{1}{3}\frac{\mathrm{d[H_2]}}{\mathrm{dt}} = \frac{1}{2}\frac{\mathrm{d(NH_3]}}{\mathrm{dt}}$

Rate of disappearance of $\mathrm{N_2} = \frac{1}{2}$ **the rate of formation of** $\mathrm{NH_3} = 1 \times 10^{-4}$

Rate of disappearance of $\mathrm{H_2} = 3/2$ **the rate of formation of** $\mathrm{NH_3} = 3 \times 10^{-4}$

Q.7 Answer:

$\mathrm{r = k[A][B]}$

Solution:

From slow reaction

Rate $= \mathrm{k[A][B]}$

Q.8 Answer:

the rate of formation of Z is half of the rate of disappearance of X

Solution:

The given reaction is : $\mathrm{2X + Y \longrightarrow Z}$

$-\frac{\mathrm{d[X]}}{\mathrm{2dt}} = \frac{\mathrm{d[Z]}}{\mathrm{dt}}$

$\therefore$ **Rate of formation of** Z **is half of the rate of disappearance of** X**.**

Q.9 Answer:

2×10^{-4}

Solution:

R.O.D of A $= \frac{\mathrm{d[A]}}{\mathrm{dt}}$

Given, $\mathrm{[A]_{initial}} = 0.50\mathrm{M}$

$\mathrm{[A]_{final}} = 0.38\mathrm{M}$

$\mathrm{dt} = 10\ \mathrm{min} = 600\mathrm{sec}$

$\mathrm{d[A]} = 0.12$

Rate $= \frac{0.12}{600} = 2 \times 10^{-4}\mathrm{s^{-1}}$.

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Q.10 Answer:

does not affect equilibrium constant of the reaction

Solution:

A catalyst affects equally both forward and backward reactions, therefore it does not affect equilibrium constant of reaction.

Q.11 Answer:

does not participate in the reaction but alters the rate of reaction

Solution:

A catalyst does not take part in chemical reaction but it alters the rate of reaction.

Q.12 Answer:

$\mathrm{r}=\mathrm{K}[\text{ sugar }]^{1}[\mathrm{H}+]^{0}$

Solution:

At pH_{5}, the half-life is independent of the concentration of sugar. Hence, the reaction is of first order in sugar.

The rate law expression is rate $=\mathrm{K}[$ **sugar** $]^{1}\left[\mathrm{H}^{+}\right]^{\mathrm{m}}$.

The relationship between half life and $\left[\mathrm{H}^{+}\right]$**is** $\mathrm{t}_{1/2} \alpha\left[\mathrm{H}^{+}\right]^{1-\mathrm{m}}$

$500 \alpha\left[10^{-5}\right]^{1-\mathrm{m}}$

$50 \alpha\left[10^{-6}\right]^{1-\mathrm{m}}$

$10 \alpha(10)^{1-\mathrm{m}}$

$1-\mathrm{m}=1$

$\mathrm{m}=0$

Hence, the rate law expression is $\mathrm{r}=\mathrm{k}[\mathrm{sugar}]\left[\mathrm{H}^{+}\right]^{0}$.

Q.13 Answer:

1 and 2

Solution:

$2.5 \times 10^{-3}=\mathrm{K}[0.5]^{\alpha}[1.0]^{\mathrm{b}}$ **...(1)**

$5 \times 10^{-3}=\mathrm{K}[1.0]^{\mathrm{a}}[1.0]^{\mathrm{b}}$**...(2)**

$1 \times 10^{-2}=\mathrm{K}[0.5]^{\mathrm{a}}[2.0]^{\mathrm{b}}$**...(3)**

Dividing equation (1) and (2)

$\frac{1}{2}=\left[\frac{1}{2}\right]^{\alpha}$

hence $\mathrm{a}=1$

Dividing equation (1) and (3)

$\frac{2.5 \times 10^{-3}}{1 \times 10^{-2}}=\left(\frac{1.0}{2.0}\right)^{\beta}$

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$\frac{1}{4} = \left(\frac{1}{2}\right)^{\beta}$
$b = 2$

Q.14 Answer:

the spontaneity of the reaction

Solution:

Reaction rates are determined by all except the spontaneity of the reaction. All the other factors influence reaction rate because for a reaction to happen particles must collide with sufficient energy and in the right orientation. The energy which is required is termed the activation energy. Many spontaneous reactions like the combustion of ethanol do not happen at observable rates under standard conditions because the particles do not have energy which matches or exceeds the activation energy. All the other factors determine, reaction rates.

Q.15 Answer:

$-\frac{\mathrm{d[A]}}{\mathrm{dt}} = \frac{1}{4}\frac{\mathrm{d[B]}}{\mathrm{dt}}$

Solution:

Rate of reaction
$= \frac{-d[A]}{\frac{1}{2}dt} = \frac{dB}{2dt}$ **or** $\frac{-d[A]}{dt} = \frac{1}{4}\frac{d[B]}{dt}$

Q.16 Answer:

May or may not have same value depending upon the stoichiometric coefficient of reactant (or product) in the balanced chemical equation

Solution:

The rate of reaction and rate of appearance (or disappearance) of a reactant (or product) are different quantities unless its stoichiometric number in chemical equation is one.

Q.17 Answer:

$\frac{0.056}{3}\ \mathrm{L\ mol^{-1}\ s^{-1}}$

Solution:

$-\frac{1}{3}\frac{d\left[\mathrm{BrO^-}\right]}{dt} = \frac{d\left[\mathrm{BrO_3^-}\right]}{dt} = +\frac{1}{3}\frac{d\left[\mathrm{Br^-}\right]}{dt}$
So, Rate $= 1/3 \times 0.056$

Q.18 Answer:

$\frac{2.303}{10}\log\frac{5}{4}$

Solution:

Reactants and products are in gaseous state thus,
$\mathrm{A(g) \rightarrow B(g) + C(g)}$

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at $t = \quad 0 \quad \phi_I \quad 0 \quad 0$

at $t = t \quad (p_i - x) \quad x \quad x$

total pressure at time

$t = p_t = p_i - x + x + x = p_i + x$

$\therefore x = (p_t - p_i)$

$(a - x) = p_i - (p_t - p_i)$

$= 2p_i - p_t$

Given $p_i = a = 100$ mm

$p_t = 120$ mm

$\therefore (a - x) = 2p_i - p_t = 80$

$\therefore k = \frac{2.303}{10} \log \frac{100}{80}$

Q.19 Answer:

8dps, 10 min

Solution:

There is error of 3 dps

$\therefore C_0 = 20$dps

$C_t = 10$dps

Thus, half-life $= 10$ **minutes**

In next 10 minutes $C_t = 5$ **dps**

$\therefore$ **recorded value with error** $= 8$dps

Q.20 Answer:

1

Solution:

Rate of the given reaction is doubled when concentration of 'A**' is doubled and it is quadrupled when concentrations of '**A**' and '**B**' are raised four times.**

Let the rate, $R = k \quad [A]^x \quad [B]^y \qquad \ldots i$

$\therefore$ **When concentration of** A **is doubled, the new rate** R_1 **will be**

$R_1 = k[2A]^x[B]^y \qquad \ldots ii$

But $R_1 = 2R$

$\therefore \quad k[2A]^x[B]^y = 2k[A]^x[B]^y$ **.....[From (i) and (ii)]**

$\therefore \quad k2^x[A]^x[B]^y = 2k[A]^x[B]^y$

$\therefore 2^x = 2$ **and thus** $x = 1$

Similarly, when concentrations of both A **and** B**are quadrupled, the new rate** R_2 **will be**

$R_2 = k[4A]^x[4B]^y \qquad \ldots iii$

But $R_2 = 4R$

$\therefore \quad k[4A]^x[4B]^y = 4k[A]^x[B]^y$

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.....[From(i)and (iii)]

$k \quad 4^x[A]^x 4^y[B]^y = 4k[A]^x[B]^y$

$\therefore \quad 4^x \cdot 4^y = 4$

$\therefore \quad 4 \cdot 4^y = 4 \quad (\because x = 1)$

$\therefore \quad 4^y = 1$ **and thus** $y = 0$

Substituting the values of x **and** y **in** i**, the rate expression for the givcn rcaction is**

rate, $R = k[A]^1[B]^0 \quad \therefore \quad R = k[A]$

Thus, the overall order of the given reaction is 1.

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